

31/07/25

Nano Device Engineering (EC 552)

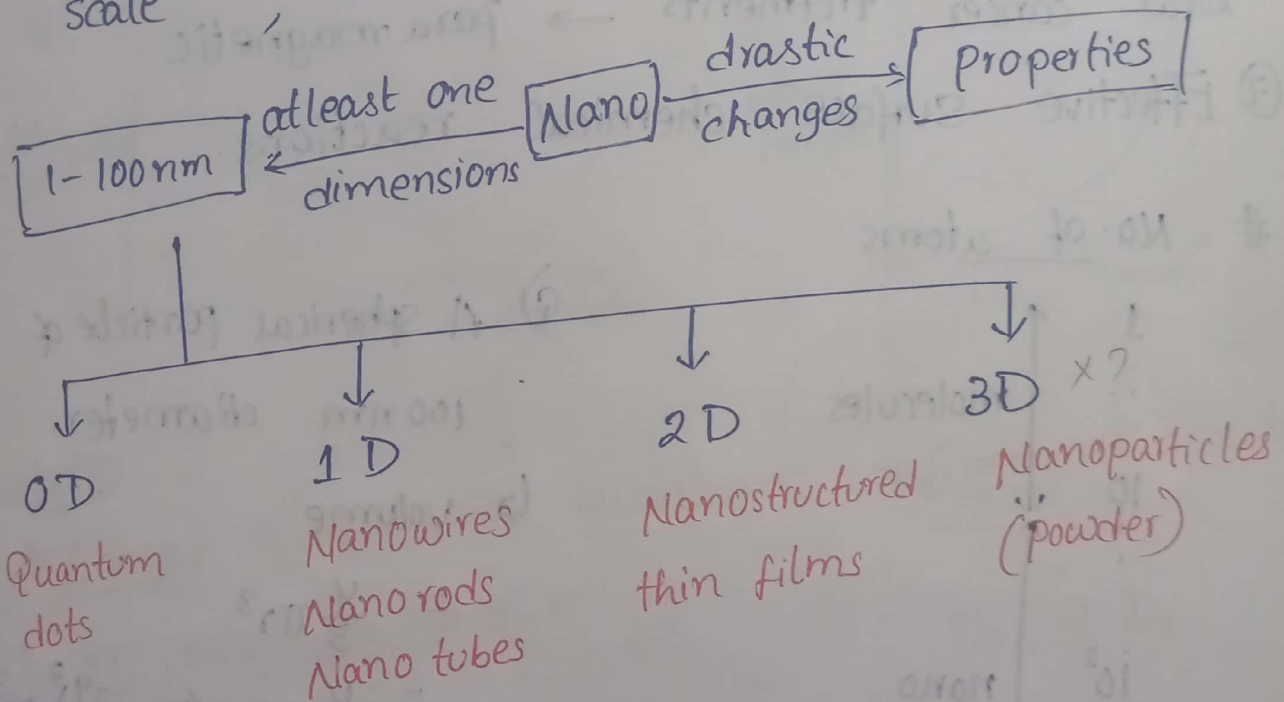
PC RAM :

Phase change Random Access memory.

↳ phase of the material gets changed.

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* At least one of the dimensions is in "nano-scale" then it is a nano device.

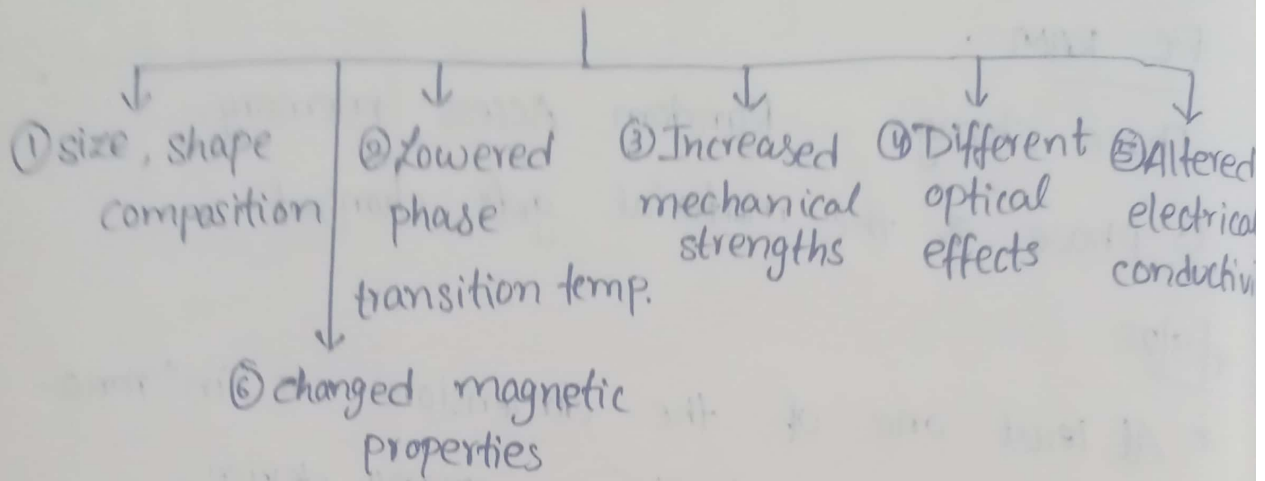


Properties

- ↳ Increased surface to volume ratio
- ↳ High % of atoms/molecules at the surface.
- ↳ Dominant surface forces
- ↳ Plasmonic resonance effects.
- ↳ Confined energy states [quantum confinement]
- ↳ Drastic physical, chemical, Biological changes.

⑥

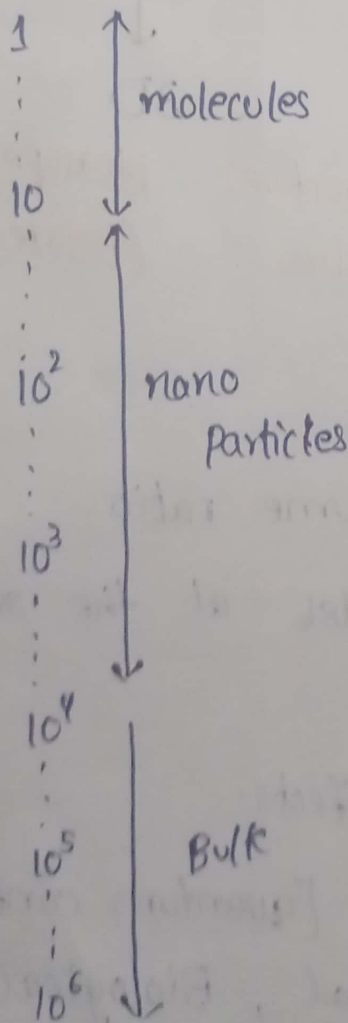
Property 6



* Local cancer treatments → Paramagnetic

⑦ Effective surface chemical reactions.

No. of atoms



Q) A spherical particle of 100 nm diameter.

↳ volume

$$= \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \pi \times (50 \times 10^{-9})^3 \text{ m}^3$$

$$V = 5.24 \times 10^{-22} \text{ m}^3$$

↳ Surface area = $4\pi r^2$

$$SA = 3.14 \times 10^{-14} \text{ m}^2$$

$$\boxed{\frac{SA}{V} = \frac{3.14 \times 10^{-14}}{5.24 \times 10^{-22}}} \quad \text{nano particle}$$

↳ If $d = 100 \text{ m} \Rightarrow r = 50 \text{ m}$

$$V = \frac{4}{3} \times \pi \times 50^3 = 5.23 \times 10^5 \text{ m}^3$$

$$SA = 4 \times \pi \times 50^2 = 3.14 \times 10^4 \text{ m}^2$$

$$\boxed{\frac{SA}{V} = 0.06 \text{ units}} \quad \text{Bulk particle}$$

∴ Nano particle is surface effective.

$\lambda = 100 \text{ nm}$

$$E = \frac{hc}{\lambda} = \frac{1240}{\lambda}$$

$$= 1.24 \times 10^{10} \text{ units}$$

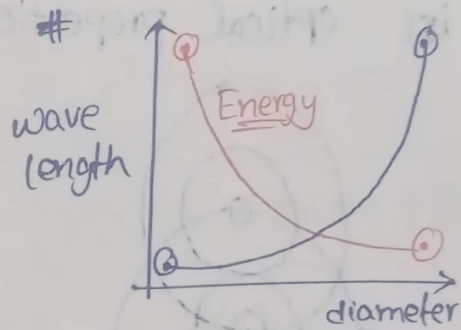
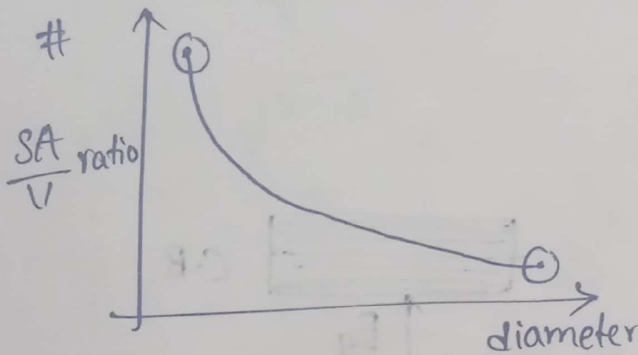
not detectable

$\lambda = 100 \text{ m}$

$$E = \frac{hc}{\lambda} = \frac{1240}{\lambda}$$

$$= 12.4 \text{ units}$$

radio region



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Energy levels and quantum confinement effects in nanostuctures :-

3D $\rightarrow$ 0D

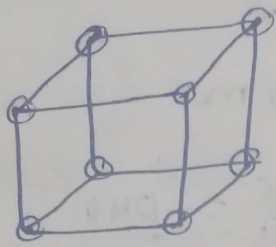
$D_f + D_c = 3$

↙ Degree of freedom ↘ Degree of confinement

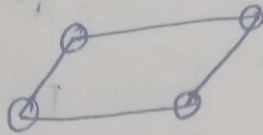
	D_f	D_c
3D	3	0
2D	2	1
1D	1	2
0D	0	3

How will the particles move?

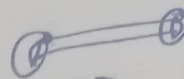
* Reducing the no. of particles on the surface gives the confinement from 3D to 2D, 2D to 1D, ...



3D



2D

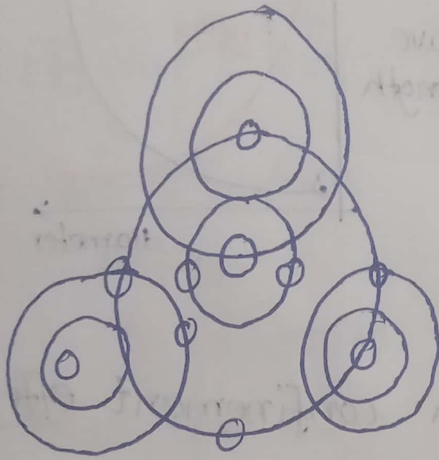


1D

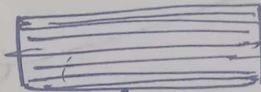


0D

In case of nano devices, Band Gap Engineering is very important as it plays an imp. role in optical properties,



10^{23} atom/cc



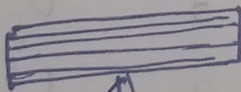
C-B

E_g

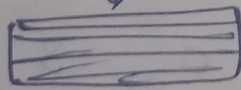


V-B

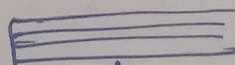
3D



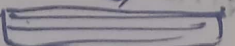
E_g



2D



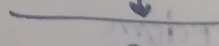
E_g



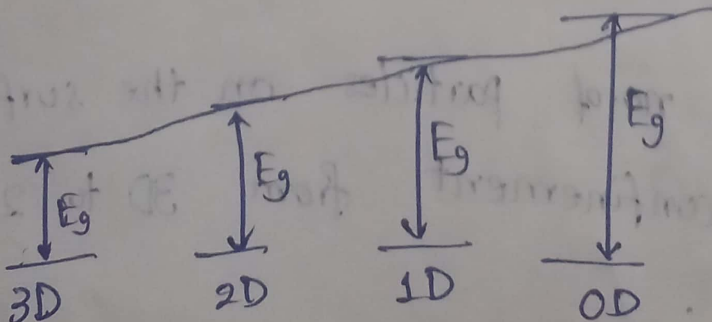
1D



E_g



0D



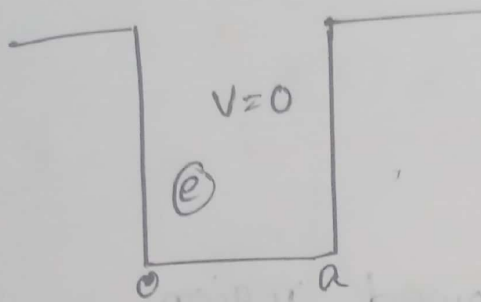
3D

2D

1D

0D

In case of 1D



Schrodinger's eqⁿ

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V) \psi(x) = 0$$

Inside the 1D well $\Rightarrow V=0$

$$\frac{d^2\psi}{dx^2} + \left(\frac{2mE}{\hbar^2} \right) \psi(x) = 0$$

$\xrightarrow{\quad \quad \quad} k^2$

$$k^2 = \frac{2mE}{\hbar^2}$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$\frac{d^2\psi}{dx^2} + k^2 \psi(x) = 0$$

Solⁿ $\rightarrow \psi(x) = A \sin kx + B \cos kx$

If $\rightarrow x=0 \Rightarrow \psi(x) = 0$

$$\psi(0) = A \sin k(0) + B \cos k(0)$$

$\xrightarrow{\quad \quad \quad} 1$

$$0 = 0 + B(1)$$

$$B = 0$$

$$\psi(x) = A \sin kx$$

If $\psi(x=a) = 0$ for $x=a$

$$\psi(a) = A \sin ka$$

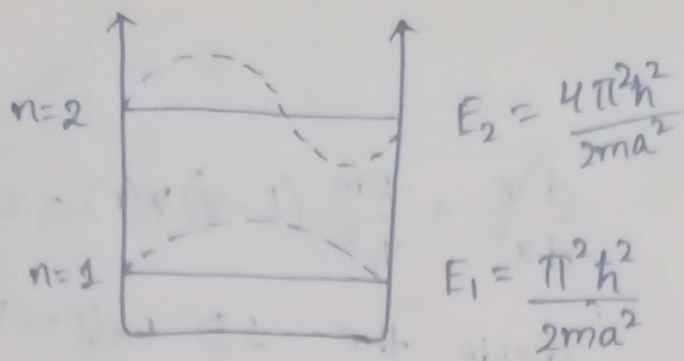
$$0 = A \sin ka \quad A \neq 0$$

$$\sin ka = 0 \Rightarrow ka = n\pi \Rightarrow k = \frac{n\pi}{a}$$

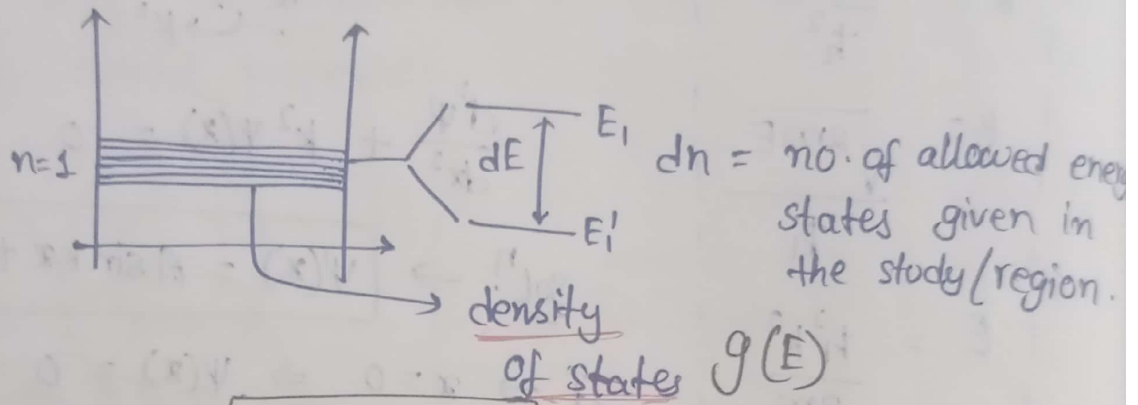
$$k^2 = \frac{n^2 \pi^2}{a^2}$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

$\rightarrow$ Energy of particle in 1D well



* no. of states in the allowed region



$$g(E) = \frac{dn}{dE}$$

$$dE = \frac{2n dn \pi^2 \hbar^2}{2ma^2}$$

$$\left(\because E = \frac{n^2 \pi^2 \hbar^2}{2ma^2} \right)$$

$$\frac{dE}{dn} = \frac{n \pi^2 \hbar^2}{ma^2}$$

$$g(E) = \frac{dn}{dE} = \frac{ma^2}{n \pi^2 \hbar^2}$$

$$\frac{dn}{dE} = \frac{ma^2}{\pi^2 \hbar^2} \frac{\pi \hbar}{a \sqrt{2mE}}$$

$$\left(\because k^2 = \frac{n^2 \pi^2}{a^2} \right)$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$n^2 = \frac{2ma^2 E}{\pi^2 \hbar^2}$$

$$n = \frac{a}{\pi \hbar} \sqrt{2mE}$$

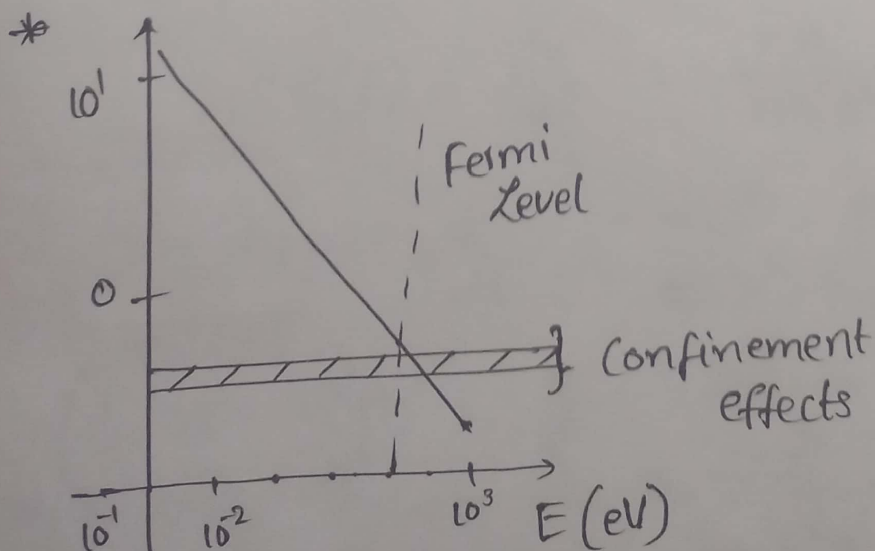
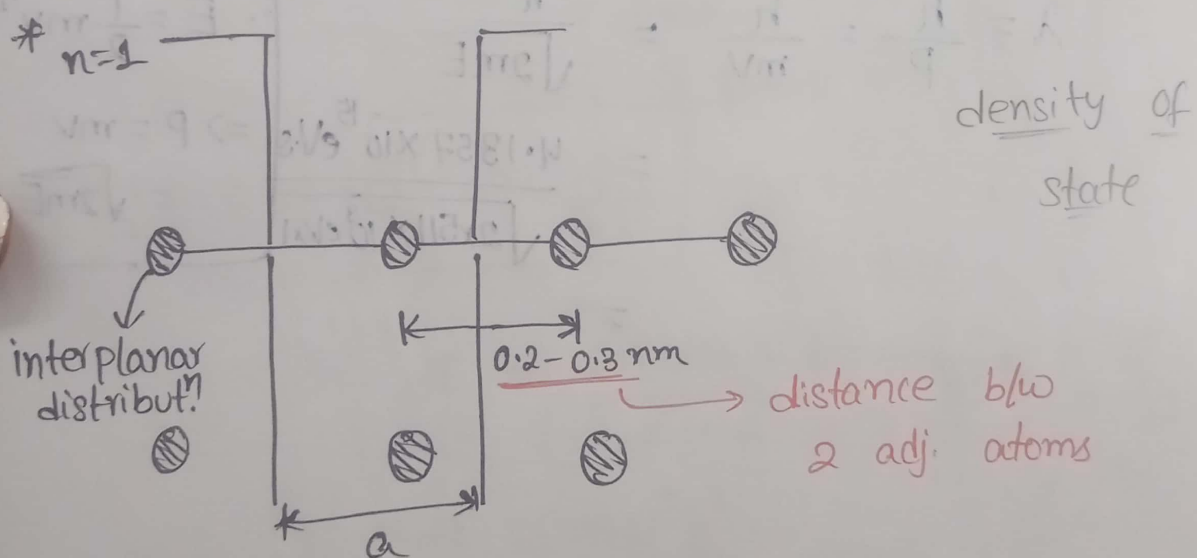
#

	KE	λ	a
electron	1 eV	1.22 nm	0.613 nm
	5 eV	0.55 nm	0.274 nm

* When wavelength (λ) is comparable to barrier constant (a) i.e. $\lambda \sim a$, only then quantum confinement takes place.

$$\lambda \sim a$$

De-Broglie's
wave length



Density of States

$$\psi(x) = A \sin kx + B \cos kx$$

$$x=0 \Rightarrow A \sin k(0) + B \cos k(0) = 0$$

$$\psi(x)=0 \Rightarrow B=0$$

$$x=a \Rightarrow A \sin$$

Density of States:

1D Dos

$$g(E) \propto \frac{dN}{dE} \propto E^{-1/2}$$

$$k = \frac{n\pi}{a}$$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

$$\text{no. of states} = \frac{\text{Vol. of sphere}}{\text{Vol. of single cell}}$$

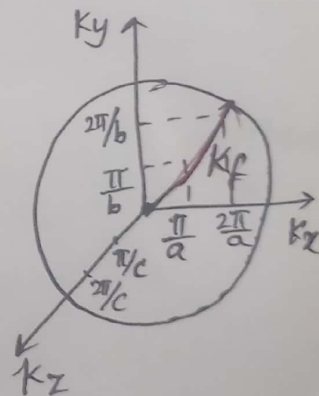
$$= \frac{\frac{4}{3} \pi k_f^3}{\left(\frac{2\pi}{a}\right)^3} \quad (\text{for } a=b=c)$$

$$= \frac{1}{6} \cdot \frac{k_f^3 a^3}{\pi^2}$$

$$n = \frac{a^3 k_f^3}{6\pi^2}$$

$$n = \frac{a^3 (2mE)^{3/2}}{6\hbar^3 \pi^2}$$

$$\frac{dn}{dE} = \frac{a^3 (2m)^{3/2} \cdot \frac{3}{2} E^{1/2}}{6\hbar^3 \pi^2}$$



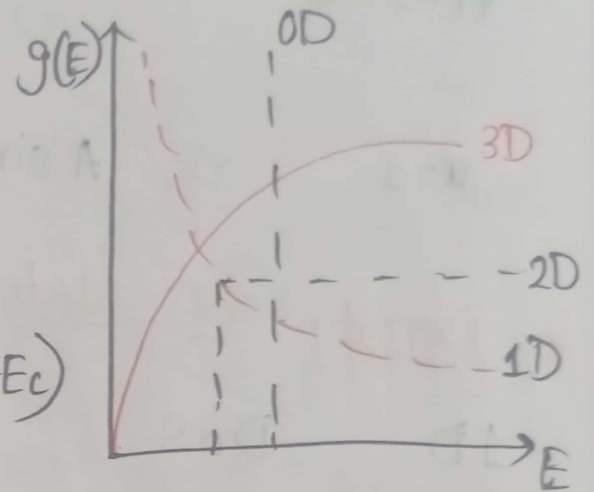
$$\frac{dn}{dE} = \frac{a^3 (2m)^{3/2} \cdot \frac{3}{2} \cdot E^{1/2}}{6h^3 \pi^2}$$

$$g(E) = \frac{dn}{dE} = \frac{a^3 \cdot m^{3/2} \cdot E^{1/2} \cdot \sqrt{2}}{2 \cdot \pi^2 h^3} \rightarrow \text{density of states}$$

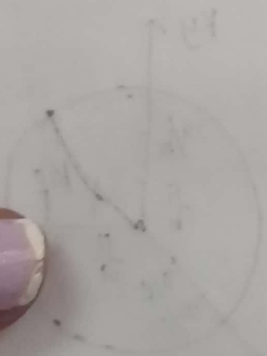
3D case $g(E) \propto E^{1/2}$

For 2D $\rightarrow g(E) \propto E^0$

for 0D $\rightarrow g(E) \propto \delta(E - E_c)$



For 2D and 0D $\frac{dn}{dE} = 0$



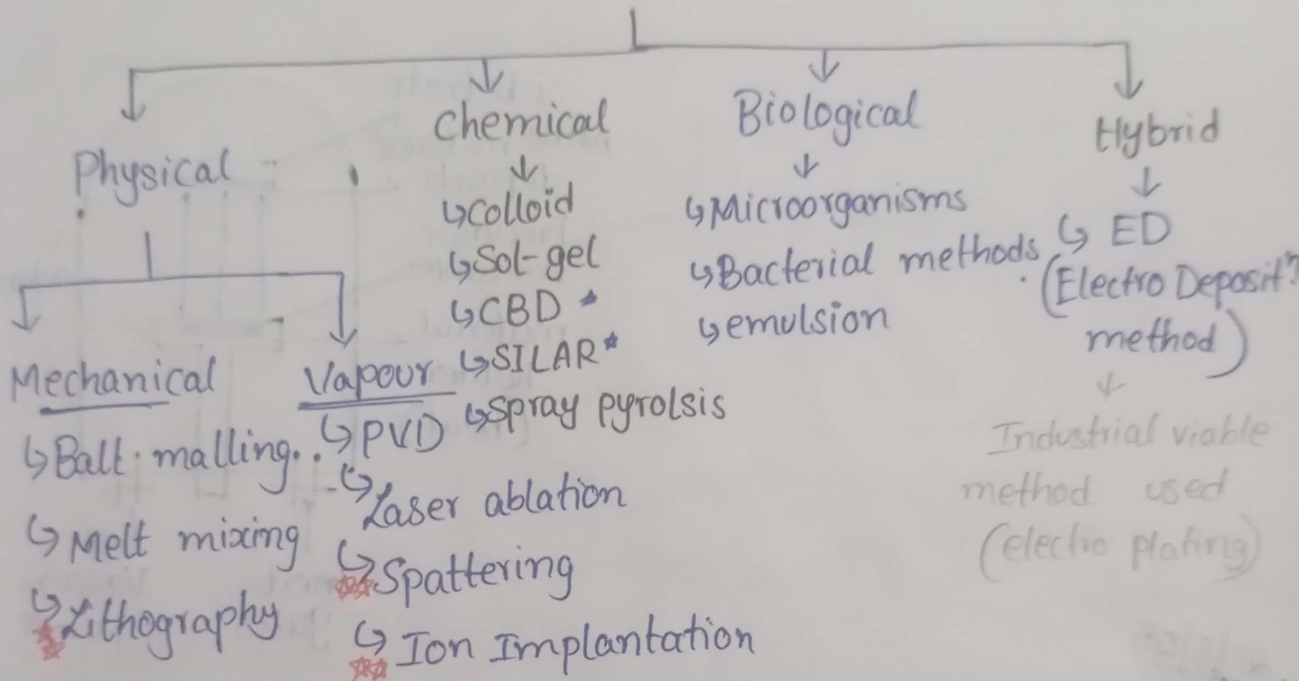
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Synthesis Methods

SILAR?

Top - Bottom Approach

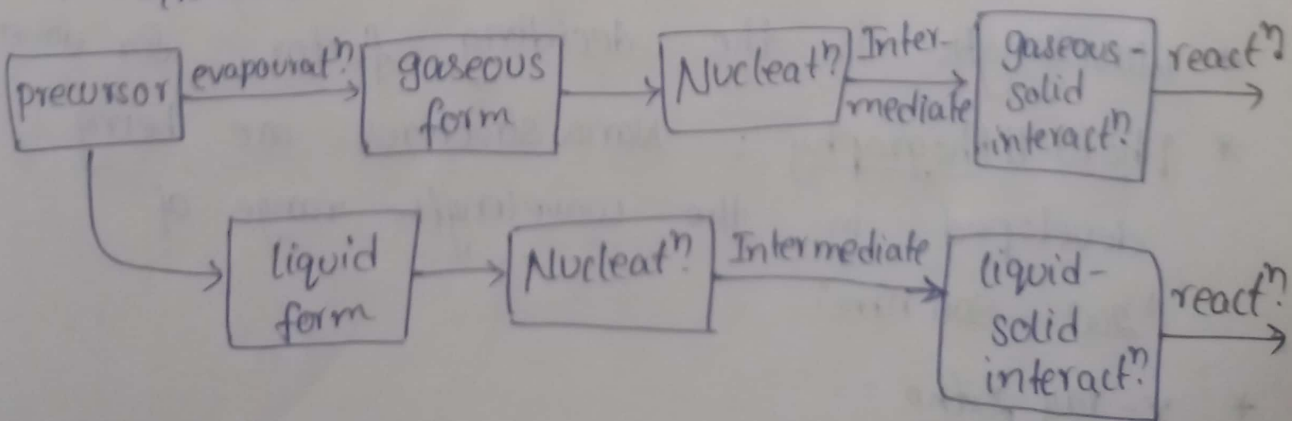
Bottom - up Approach

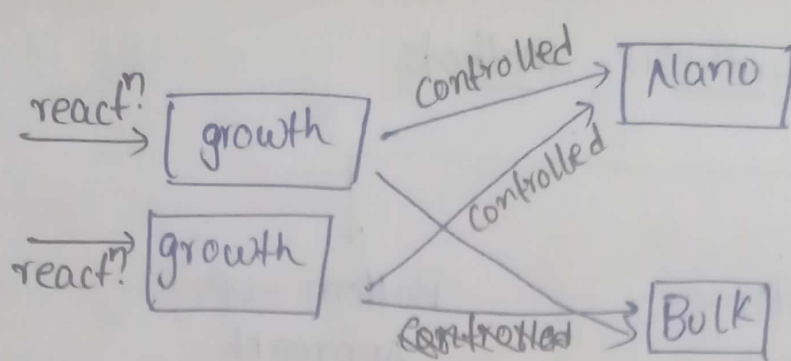


Principles / mechanisms used in Synthesis Methods

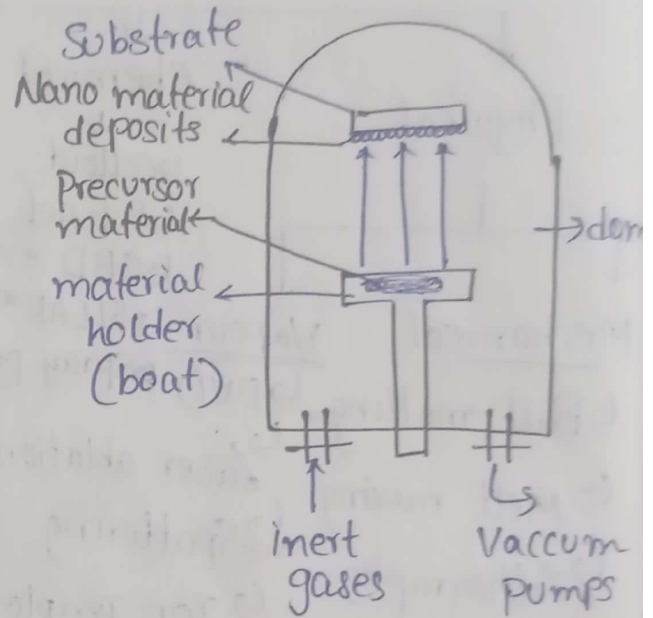
for building nano structures:

- ↳ Precursor, evaporation/liquid/gaseous
- ↳ Nucleation
- ↳ Growth





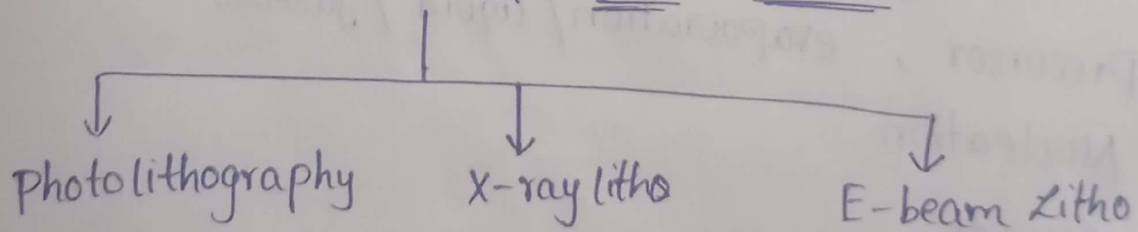
Thermal evaporation method:



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Lithography Process : → for devising Nanostructure

↳ Transferring of geometrical patterns to develop nano structures.



* Wavelength is the deciding factor for dimension

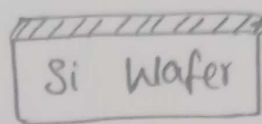
* Photo lithography : Nano structures are being developed in the wavelength range of "200 - 800 nm".

* X-ray Litho :

Steps:

① Coating the photoresist on Silicon wafer.

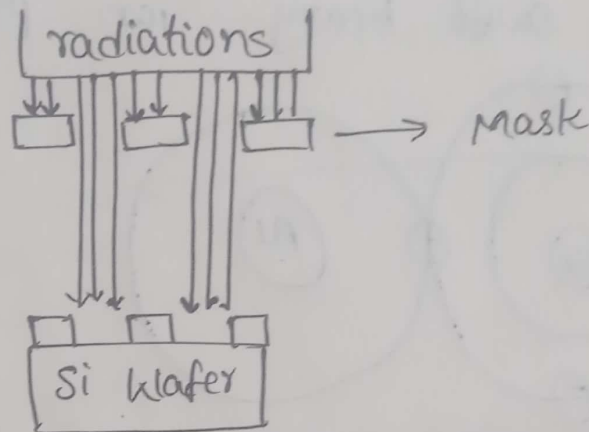
spin coater



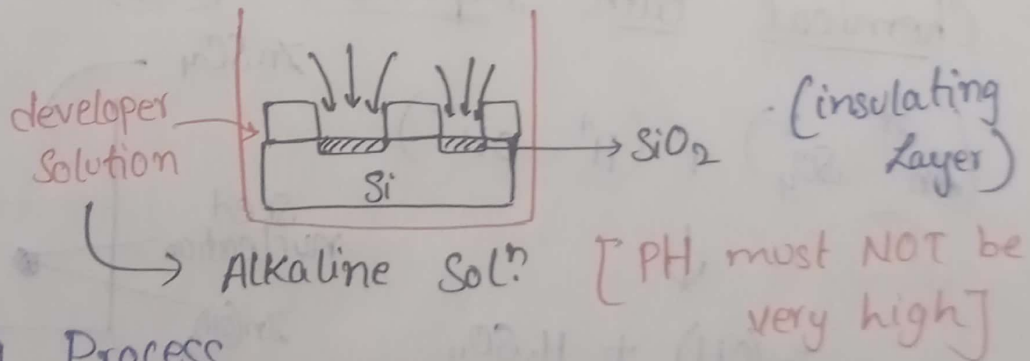
photoresist*

photo sensitive polymeric material

② Exposure to radiations.

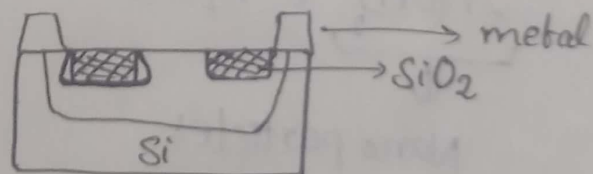


③ Use developer for developing the necessary pattern.



④ Etching Process

→ HF



* Microfabrication of semi conductors → MEMS

* Nanofabrication → NEMS

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* Ion Implantation

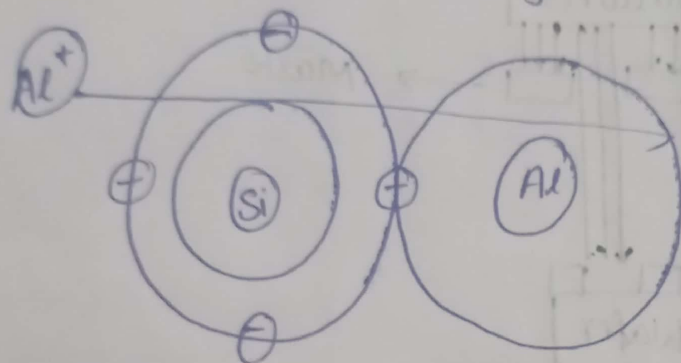
→ Centre (in Delhi)

Ion interacts with

- ① electronic energy
- ② Nuclear energy

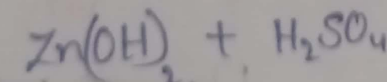
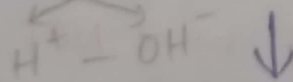
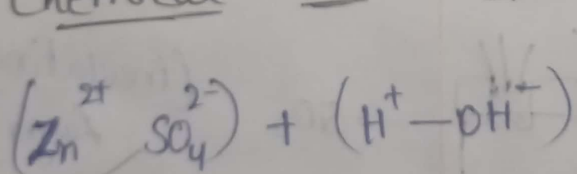
* swift energy is high

→ swift heavy ion process

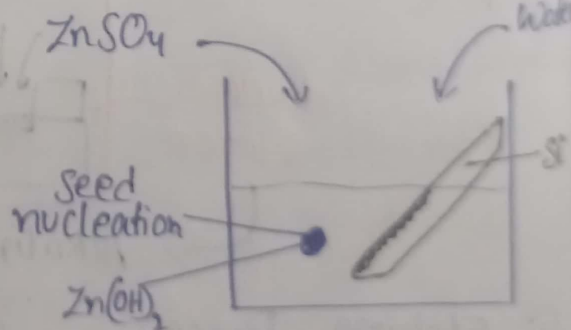


* CBD is the easiest method used for doping

* Chemical Bath Deposition (CBD):



↓
Nano particles



for controlling the growth of Zn(OH)_2

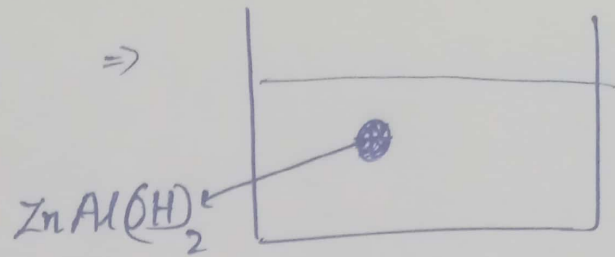
-
- ① Volumetric control of chemicals
 - ② Temperature control

⑥ 1:2

① ZnSO_4

② Al_2SO_4

$\Rightarrow$



$\hookrightarrow$ Doped with Al

* Crude method for doping is CBD. But it is not precise.

* Ion implantation is precise process.